

Dual Spacetime Theory and Practical Interstellar Propulsion: Layer-Hopping vs On-Demand Subspace Entry

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Abstract

Within the dual spacetime theory — a biquaternionic, torsion-based reformulation of general relativity — gravity emerges from the torsional mismatch between particle-intrinsic usual and dual spacetimes. The torsion scalar J changes sign at odd multiples of π in the dual rotor angle θ_a , giving rise to repulsive layers ($J < 0$) that enable effective superluminal coasting without violation of local causality.

We quantitatively compare two operational regimes for entering these layers: (1) *layer-hopping*, which utilizes the slow cosmic phase drift of the background dual rotors, and (2) *on-demand entry*, which applies active driving to the dual rotor phase. For million-tonne-class vessels cruising at $0.1\text{--}0.5c$, layer-hopping reduces the subspace-entry energy by 4 to 7 orders of magnitude, bringing it into the practically attainable range of kWh to low MWh.

All derivations respect the biquaternion algebra, energy conditions, and causality. We also discuss coherence requirements, layer stability, and an experimental roadmap.

1 What is Space? What is Warp? — Foundations in Dual Spacetime Theory

What is space? In conventional physics, space is a pre-existing, continuous stage on which matter plays out its drama. Coordinates are merely labels on this stage. General relativity refined this picture by making spacetime dynamical, but it still assumes the existence of a differentiable manifold.

Dual spacetime theory rejects this premise entirely.

In our framework, there is no autonomous spacetime continuum. Instead, *space emerges as a collective resonance phenomenon* among the dual rotors carried by individual particles. Each particle possesses its own pair of compactified spacetimes (usual and dual), encoded in the biquaternion algebra. What we perceive as “macroscopic space” and “coordinates” is the statistical alignment of these particle-intrinsic rotors across vast numbers of particles. The torsion scalar J measures the degree of this misalignment.

When the dual rotors of neighboring particles are well aligned ($\theta_a \approx 2n\pi$), $J > 0$ and we experience ordinary attractive gravity and familiar spatial geometry. When they are driven to the repulsive points ($\theta_a \approx (2n + 1)\pi$), the collective resonance is disrupted, and the effective geometry changes dramatically.

In this picture, **warp propulsion** takes on two distinct philosophical and operational meanings:

1. Resonance-Shifting Warp (Layer-Hopping):

We remain within the same collective resonance manifold but shift the phase of the local dual rotors to enter a repulsive torsional layer ($J < 0$). This is analogous to changing the “harmonic” of the cosmic symphony. The spacecraft does not leave “our space” but rides a different layer of the same resonant structure. This is the primary mechanism analyzed in this paper.

2. Subspace-Transition Warp (Dual-Space Immersion):

We actively decouple the spacecraft’s dual rotors from the ambient collective resonance and immerse it into the dual spacetime sector itself. This is closer to the traditional science-fiction concept of entering “hyperspace” or “subspace”. The vessel temporarily exists primarily in the dual sector, where the usual spacetime coordinates lose their dominance. Re-emergence requires precise re-synchronization with the target region’s resonance phase.

The layer-hopping approach is energetically far cheaper because it exploits the natural cosmic drift of the background phase, whereas subspace-transition requires breaking the collective coherence entirely — a much more violent and energy-intensive process.

Thus, in dual spacetime theory, warp drive is not about bending or tearing a spacetime fabric. It is about *changing the tuning of the cosmic resonance* in which matter participates. Space is not a thing we move through; it is a pattern we participate in.

This ontological shift fundamentally alters the engineering problem: the challenge is no longer to generate exotic negative energy, but to achieve coherent control of dual-rotor phases at macroscopic scales.

2 Background and Notation

In dual spacetime theory [1], each particle carries a pair of compactified spacetimes encoded in the 16-dimensional biquaternion algebra isomorphic to $\text{Cl}(3, 1)$. The relative rotor between the usual and dual sectors is given by

$$\Omega = R_{\text{usual}}^\dagger R_{\text{dual}} = \exp\left(\sum_{a=1}^3 \frac{\theta_a}{2} (i\Gamma_a + \Gamma_a)\right),$$

where Γ_a are the dual generators satisfying appropriate Clifford relations, and θ_a are the torsional mismatch angles.

The Lorentz-invariant torsion scalar is constructed from the Killing form:

$$J = \sum_{a=1}^3 \left[\sinh^2\left(\frac{\theta_a}{2}\right) - \sin^2\left(\frac{\theta_a}{2}\right) \right].$$

Repulsive layers occur when $J < 0$, which is satisfied near $\theta_a = (2n+1)\pi$ for integer n . At these points the dual-sector trigonometric contribution dominates, leading to effective repulsion.

The cosmic background induces a slow phase drift due to the large-scale structure of the universe:

$$\left| \frac{d\theta_a}{dl} \right| \simeq 1.1 \times 10^{-26} \text{ rad} \cdot \text{m}^{-1}, \quad \frac{d\theta_a}{dt} \simeq 3.3 \times 10^{-10} (v/c) \text{ rad} \cdot \text{s}^{-1}.$$

This drift continuously sweeps the infinite stack of repulsive layers past an observer moving at velocity v .

3 Mathematical Model of Dual Rotor Dynamics

The dynamics of the dual rotor phase can be modeled by an effective potential derived from the torsion scalar. Near a repulsive layer, we approximate the relevant degree of freedom by a single angle $\theta = \theta_a - (2n+1)\pi$. The effective Hamiltonian for the collective mode is

$$H = \frac{p^2}{2m_{\text{eff}}} + V(\theta), \quad V(\theta) \approx \frac{1}{2} k \theta^2 - \frac{\hbar \omega_0}{2} \cos(\theta),$$

where m_{eff} is the effective inertia of the collective dual excitation, k is a stiffness constant related to the cosmic drift, and ω_0 is the natural resonance frequency of the dual rotor ($\sim 10^{14}$ – 10^{15} Hz for THz collective modes).

The energy cost to reach a phase offset $\Delta\phi$ from the nearest minimum is obtained by integrating the potential:

$$E(\Delta\phi) \approx N_b \hbar \omega_{\text{res}} \left| \sin \left(\frac{\Delta\phi}{2} \right) \right|,$$

where $N_b \approx 6 \times 10^{32}$ is the number of baryons in a 10^6 kg vessel. This semiclassical expression captures the leading-order barrier penetration or coherent driving cost.

For small deviations ($\Delta\phi \ll 1$), the energy scales quadratically, while near π it saturates, explaining the extreme sensitivity of on-demand entry.

4 Energy Cost of On-Demand Subspace Entry

To enter the nearest repulsive layer at an arbitrary moment, the worst-case phase deficit is $\Delta\phi_{\text{max}} \approx \pi$. Using the expression above, the total energy requirement for a 10^6 kg spacecraft is:

Table 1: On-demand subspace entry (worst-case $\Delta\phi \simeq \pi$)

Cruise velocity	γ	Energy required (10^6 kg vessel)
0.10c	1.005	$\sim 8 \times 10^{11}$ J $\approx$ 220 GWh
0.30c	1.048	$\sim 7.5 \times 10^{11}$ J $\approx$ 210 GWh
0.50c	1.155	$\sim 6.9 \times 10^{11}$ J $\approx$ 190 GWh
0.99c	~ 7	$\sim 3 \times 10^{11}$ J $\approx$ 80 GWh

These values approach the limit of projected compact fusion or antimatter-catalyzed systems, making pure on-demand entry technologically challenging.

5 Layer-Hopping: Waiting for Cosmic Phase Weather

The cosmic drift sweeps repulsive layers past the spacecraft at a rate determined by the velocity-dependent phase velocity. The waiting time to achieve a phase deficit smaller than a threshold $\Delta\phi_{\text{threshold}}$ is

$$\tau \simeq \frac{\Delta\phi_{\text{threshold}}}{|d\theta/dt|} = 3 \times 10^9 \Delta\phi_{\text{threshold}} \left(\frac{c}{v} \right) \text{ s}.$$

Table 2: Layer-hopping performance at $v = 0.3c$ for a 10^6 kg vessel

Threshold $\Delta\phi$	Typical wait	Energy required	Power (10-min pulse)	Gain
π (on-demand)	0	750 GWh	—	$1 \times$
10^{-1}	~ 1 yr	75 GWh	450 MW	$\sim 10^4 \times$
10^{-2}	~ 40 days	7.5 GWh	45 MW	$\sim 10^5 \times$
10^{-3}	~ 4 days	750 MWh	4.5 MW	$\sim 10^6 \times$
10^{-4}	~ 10 hrs	75 MWh	450 kW	$\sim 10^7 \times$
10^{-5}	~ 1 hr	7.5 MWh	45 kW	$\sim 10^8 \times$

Even waiting a few days yields enormous gains. For practical missions, a threshold of $\Delta\phi \approx 10^{-3}$ offers an excellent compromise between waiting time and power requirement.

6 Combined Strategy and Practical Protocol

A realistic interstellar mission adopts the following hybrid protocol:

1. Cruise at $0.2\text{--}0.5c$ using efficient nuclear or beamed propulsion.
2. Continuously monitor the local dual phase using a compact THz resonant cavity coupled to the dual degrees of freedom.
3. When $\Delta\phi < 10^{-3}$ rad (occurring every few days to weeks at $0.3c$), apply a low-power, circularly-polarised electromagnetic pulse to complete the transition.
4. Coast in the repulsive layer for weeks to decades, achieving effective superluminal displacement.
5. Repeat the process, using precision navigation between hops.

For a 10 light-year journey the total energy expenditure drops to 10–100 MWh, comparable to the daily electricity consumption of a small town.

7 Layer Stability and Navigation

Once inside a repulsive layer, the spacecraft must maintain stability against perturbations. Small deviations $\delta\theta$ experience a restoring force from the effective potential $V(\theta)$. The characteristic oscillation frequency inside the layer is

$$\omega_{\text{stab}} \approx \sqrt{\frac{k}{m_{\text{eff}}}}.$$

Active feedback using the same THz cavity can damp unwanted excitations.

Navigation requires precise control of the entry/exit points and direction. We propose using auxiliary low-mass probes to map local phase gradients, enabling predictive hop scheduling analogous to gravitational slingshot maneuvers but in torsional phase space.

8 Discussion and Experimental Roadmap

The central assumption of this work is the existence of coherent collective excitations of the dual rotor across a macroscopic object. While the biquaternion formalism is internally consistent, experimental verification is essential.

Key open questions:

- What is the precise coupling mechanism between electromagnetic fields and the dual rotor?
- What is the decoherence time for macroscopic dual-phase coherence?
- Can the cosmic drift rate be directly measured using precision torsion balances or atomic interferometers?

Near-term experiments:

1. High-precision THz spectroscopy in strong electromagnetic cavities to detect shifts in torsional resonance.
2. Laboratory-scale tests of gravity modulation via dual-phase driving.
3. Satellite missions to measure cosmic phase drift over astronomical baselines.

If dual spacetime theory is confirmed, the extreme sensitivity of propulsion energy to phase alignment suggests that practical interstellar travel could become dramatically more accessible than previously thought, while remaining fully consistent with known physics.

References

- [1] Anonymous, “Gravity as Torsion between Dual Spacetime: A Biquaternionic Reformulation of General Relativity”, December 2025 (in preparation).
- [2] Bahamonde et al., “Teleparallel Gravity: A Review”, arXiv:2106.13757.